

TABLE OF CONTENTS

TABLE OF CONTENTS	1
1- INTRODUCTION	2
2- BIT3 DIAGNOSTICS	2
2-1 Full Host/TPS Communication Test.....	4
2-2 Host Board Only Test.....	4
2-3 Advanced Bit3 Test	4
2-4 Bit3 Diagnostic Troubleshooting Flowchart	5
REVISION HISTORY	7

1- INTRODUCTION

Beginning with Release 8.3 forward production systems, the Fiber Optic Modules have been removed from the system. The new Bit3 boards have on-board fiber optics, therefore reducing the communication path to only three pieces (i.e. Computer Bit3, TPS Bit3, and Fiber). There is a spare set of fiber in the bundle for troubleshooting. Additionally, the new Bit3 boards have proprietary loopback diagnostics that will isolate any failing component of the Bit3 system.

Note

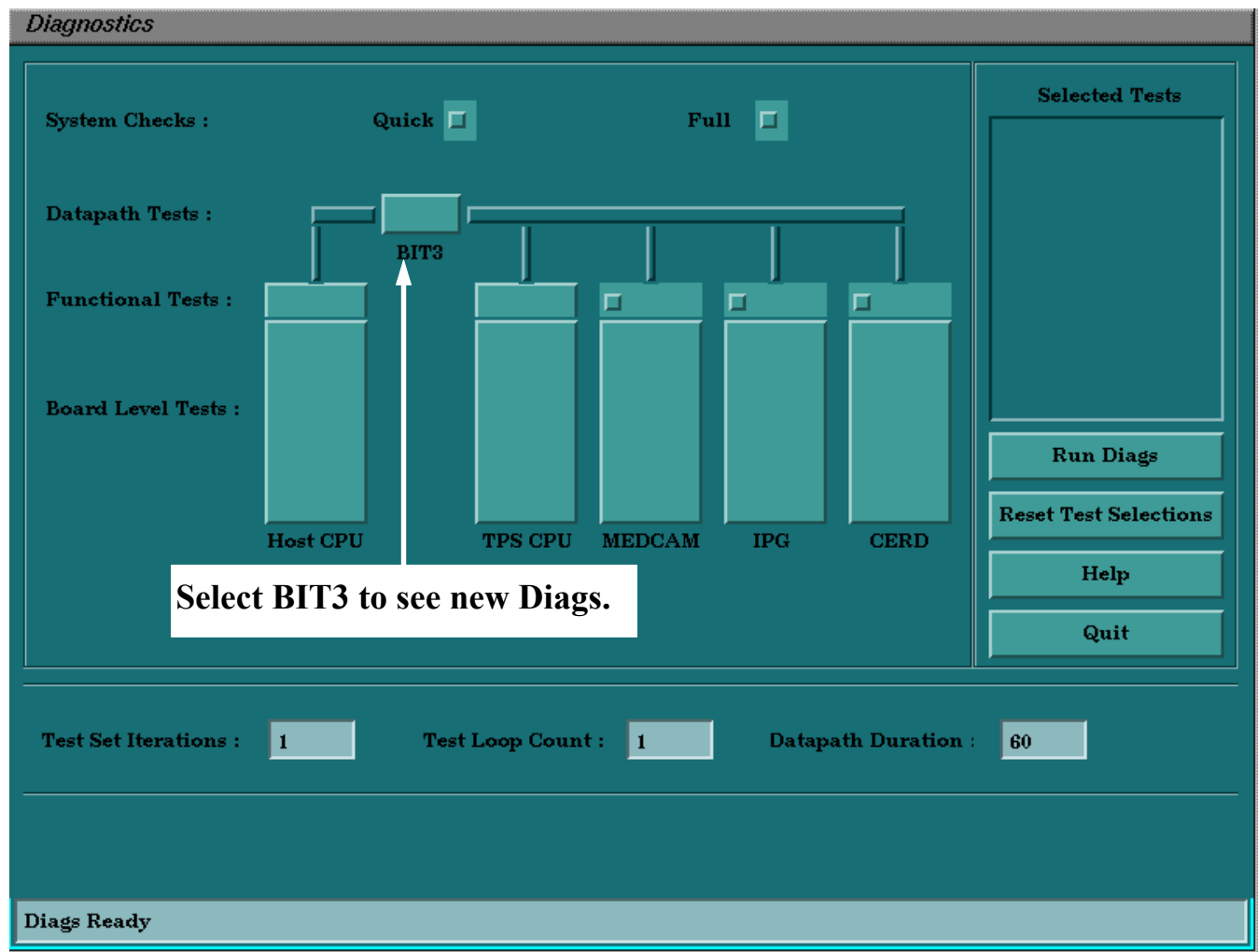
The new on-board fiber optic Bit3 boards will only work on systems with an Octane computer because of the bus structure. Indigo computers use a GIO64 bus and Octane uses a PCI bus. The new board is only compatible with the PCI bus. No on-board fiber optic Bit3 board has been designed for Indigo. The fiber optic cables from the old system are not compatible with the new system.

2- BIT3 DIAGNOSTICS

The new Bit3 board has new proprietary diagnostics. The diagnostics are started from the diagnostic GUI shown in Illustration 3-1, provided a Service Key is installed. The new diags send data across the bit3 link and verify that it makes it back to the Host.

Each buffer sent is 256 bytes long. For each run of a BIT3 diag, the following occurs:

1. Send one buffer filled with the 0xAAAAAAAA data pattern over the BIT3 link.
2. Verify the buffer received back is also filled with 0xAAAAAAAA.
3. Send one buffer filled with the 0x55555555 data pattern over the BIT3 link.
4. Verify the buffer received back is also filled with 0x55555555.



DIAGS SCREEN FOR NEW BIT3
ILLUSTRATION 2-1

Note

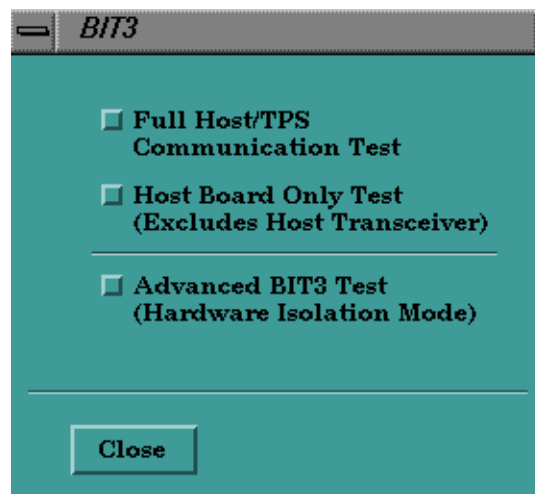
The new Bit3 Diag screen will only show up if it recognizes the system has the new Bit3 board in the Octane computer and a Service Key is installed.

The Bit3 Diags screen (Illustration 3-2) has three options to choose from.

- Full Host/TPS communication
- Host Board Only Test
- Advanced Bit3 Test

Note

Advanced Bit3 Test cannot be run with the other tests because it requires human intervention.



BIT3 DIAGS SCREEN
ILLUSTRATION 2-2

2-1 Full Host/TPS Communication Test

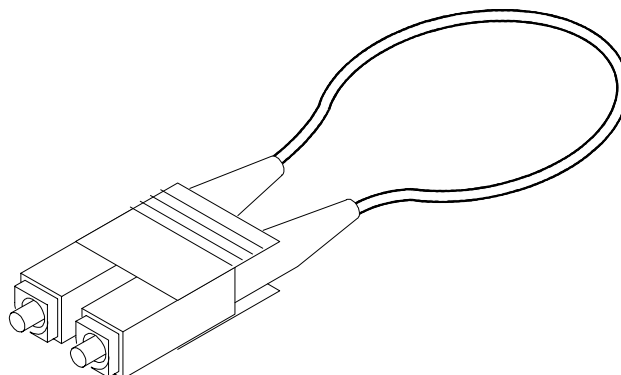
This diagnostic will test the entire Bit3 chain. This should be the first test to be run. If this test passes, increase the iterations and rerun the test. If it still passes, and you are experiencing TPS download problems revert to the traditional troubleshooting tests (i.e. ping, spray, etc.) If the Full Host/TPS test fails, run the Host Board Only test, and the Advanced Bit3 tests to isolate the failing FRU.

2-2 Host Board Only Test

The Host Board Only test will run a loopback diagnostic on the Bit3 board in the Host computer. A failure of this diagnostic is most likely the Host Bit3 board in the computer. If this diagnostic passes, however, it does not mean that the Host bit3 board is OK. This test does not test the fiber optics on the Host Bit3 board, therefore the diags can pass even though the fiber optics on the Host Bit3 board are the problem.

2-3 Advanced Bit3 Test

The Advanced BIT3 Diagnostic requires the use of a fiber optic loopback connector. There are two loopback connectors provided with the system. The TPS loopback connector is located on the BIT3 board in the TPS chassis and is labeled "Loopback". The Host loopback connector is located on the fiber optic wire harness attached to the spare fiber optic cables on the host side of the BIT3 chain. See illustration 3-3 below for identification of the loopback connector.



**BIT3 LOOPBACK CONNECTOR
ILLUSTRATION 2-3**



**ADVANCED BIT3 TEST MUST BE SELECTED BEFORE CONNECTING THE
LOOPBACK CONNECTOR OR THE SYSTEM WILL CRASH.**

Note

The Advanced BIT3 Test should only be run if the “Full TPS/Host Communication Test” fails. The loopback connectors can be used in two different locations in the BIT3 chain. Follow these steps to isolate faulty BIT3 modules and identify the failing FRU.

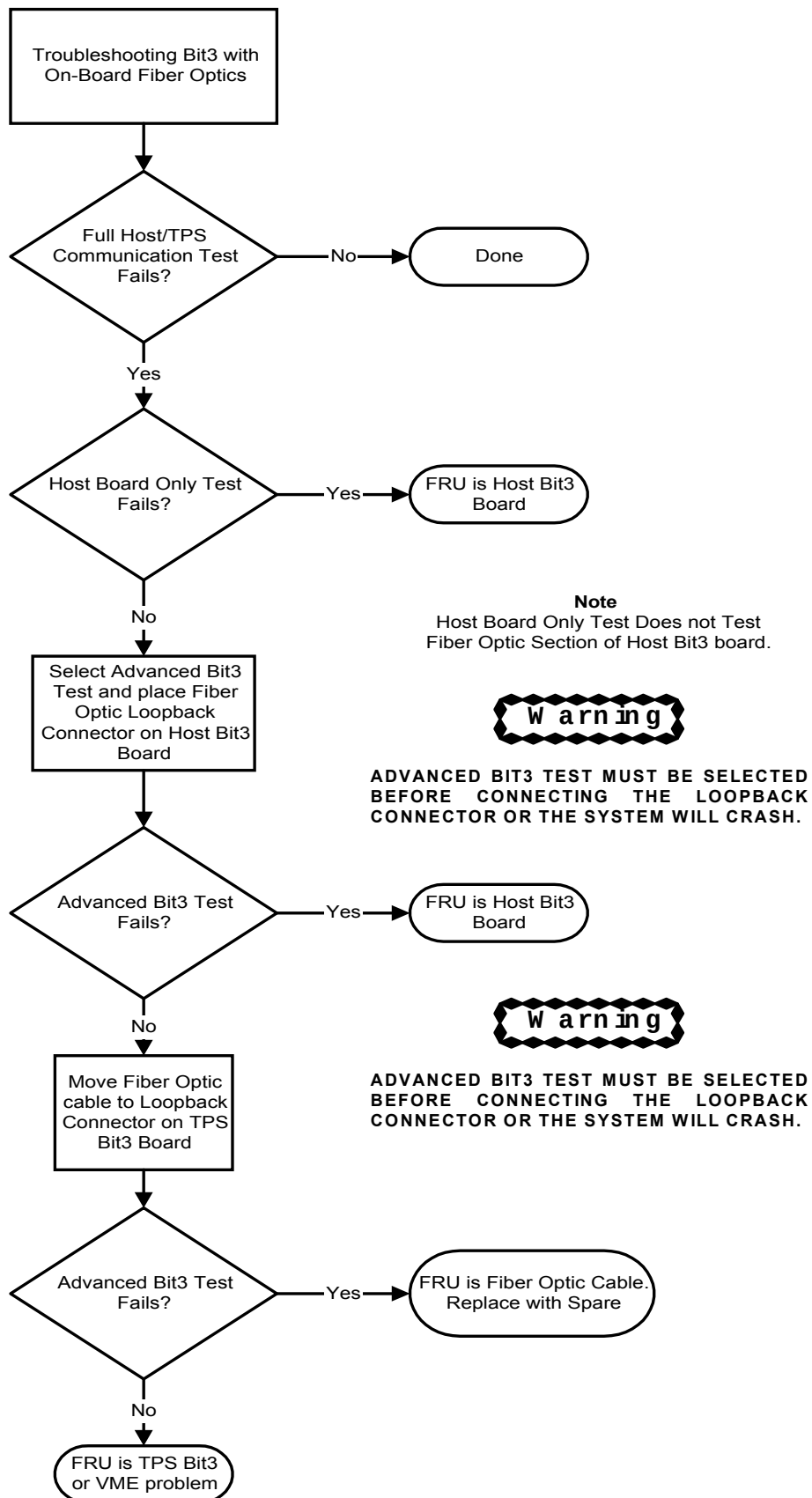
1. First, simply move the loopback connector on the Host side from the spare wire harness and attach it directly to the Host BIT3 board. Run the BIT3 Advanced Diagnostic. If the Diagnostic fails, the FRU is the Host BIT3 Board. 2235744-3. If the Diagnostic passes, restore the original configuration and continue to step 2.
2. Move the fiber optic cable from the TPS BIT3 board to the connector above it labeled “Loopback”. Run the Advanced BIT3 Diagnostic. If the Diagnostic fails, the FRU is the fiber optic cable. Use the spare cables inside the fiber optic cable harness. If the Diagnostic passes, but the “Full Host/TPS Communication Test” fails, the FRU is the TPS BIT3 board. 2235744-2.

Note

These Instructions will pop up on the system when Advanced Bit3 Test is selected.

2-4 Bit3 Diagnostic Troubleshooting Flowchart

Illustration 2-4 shows a troubleshooting flowchart for the new Bi3 diags.



BIT3 DIAGNOSTIC TROUBLESHOOTING FLOWCHART
ILLUSTRATION 2-4

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	09/29/99	R. Kaufman	Initial Release for version 8.3 software
1	10/01/99	R.Kaufman	Added Section 2-4, Flowchart
2	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.